

PanoSAMic: Panoramic Image Segmentation from SAM Feature Encoding and Dual View Fusion

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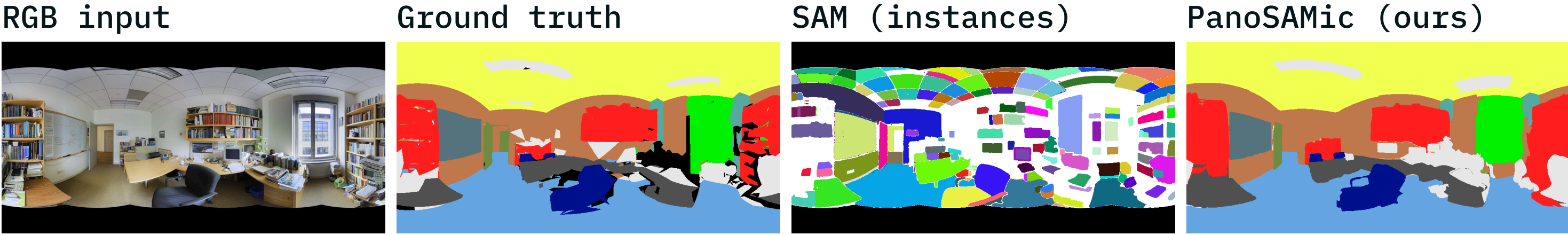
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Code, models
& demo
github.com/dfki-av/PanoSAMic



Motivation

- A single spherical sensor sees the whole scene, but the equirectangular projection distorts geometry and cuts objects at the left–right seam.
- Image foundation models are trained on perspective images: SAM leaves large panoramic regions unsegmented and carries no semantics.

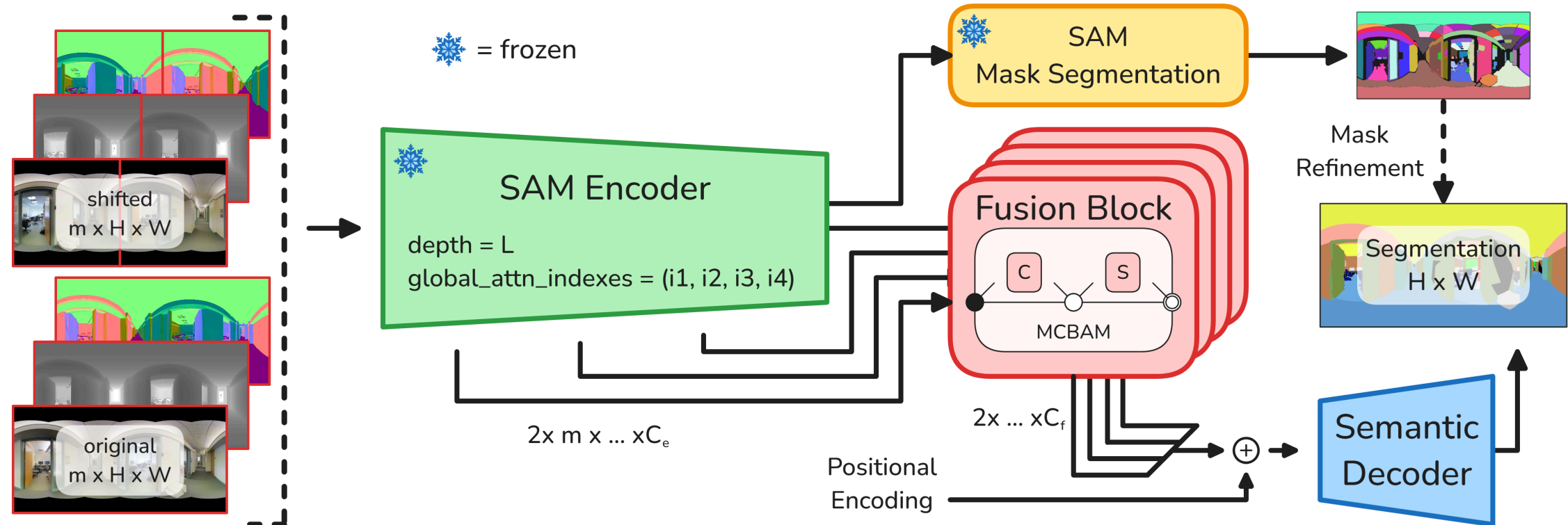


White marks what SAM leaves unsegmented; PanoSAMic reuses its pre-trained encoder.

Contributions

- The **frozen SAM encoder** in a panoramic segmentation model, extended to emit multi-stage features at **no extra parameters**.
- Dual-view fusion**: panorama and half-width-shifted copy segmented in parallel, then blended.
- Moving CBAM (MCBAM)**: sliding-window channel and spatial attention for spatio-modal fusion of RGB, depth and normals.
- State of the art** on Stanford2D3DS and Matterport3D across every modality tested.

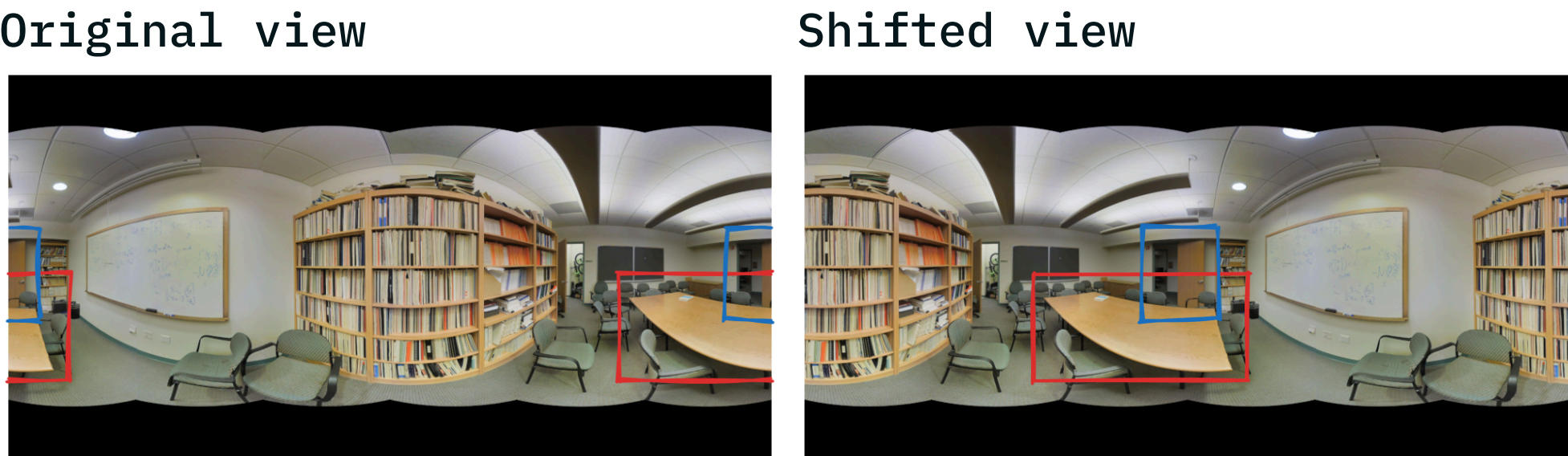
Architecture



The frozen SAM encoder emits a branch output after every global-attention layer. Per-branch MCBAM blocks fuse the modalities; SAM instance masks then refine the semantics by majority vote.

Dual-View Fusion

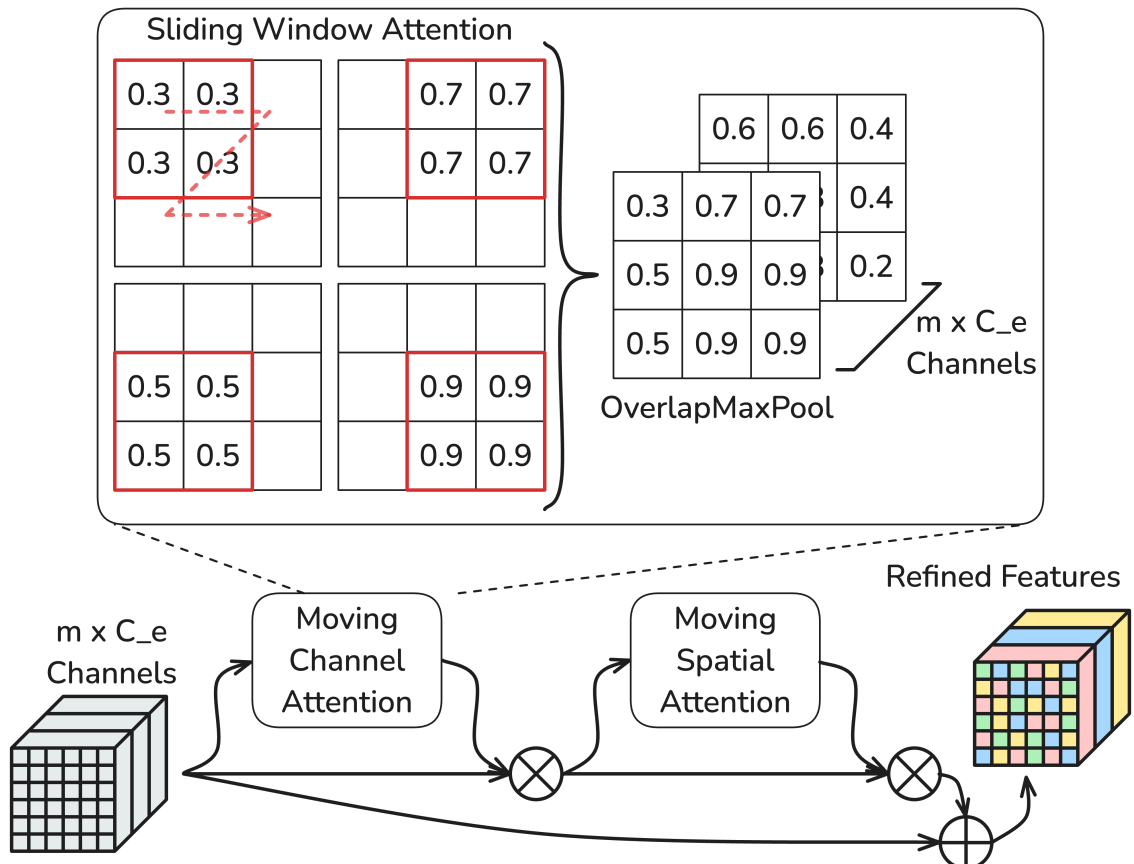
- Shift by $s = W/2$: $X_R(i, j) = X(i, (j+s) \bmod W)$. Every object is then whole in at least one view.
- Both views share the encoder as one batch; a 1-D **horizontal positional encoding**, shifted for the shifted view, aligns the feature sets.



Objects cut at the image border (red) are processed whole in the shifted view.

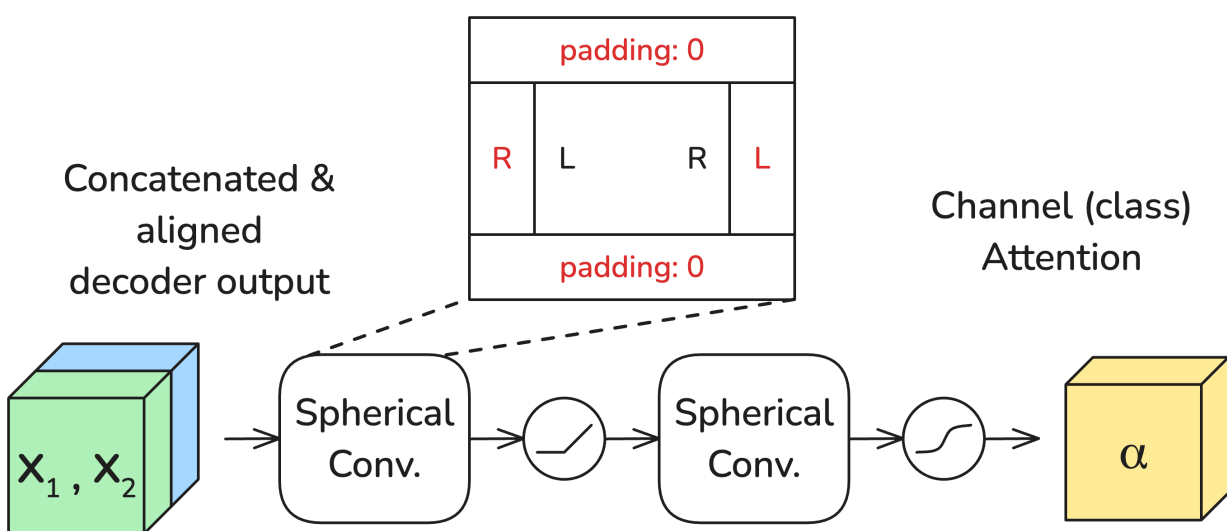
Moving CBAM

- CBAM pools *globally*: right for classification, wrong for dense prediction.
- MCBAM slides channel then spatial attention over 8×8 windows (stride 4); overlaps are max-pooled and summed through a sigmoid. On concatenated features it *selects* the best feature (and modality) per region.



Spherical View Fusion

- The realigned views are blended per class: $x_{pred} = \alpha \cdot x_1 + (1-\alpha) \cdot x_2$, with α predicted by a spherical attention block.
- Spherical convolution** pads the left border from the right and vice versa, so features interact across the full 360°.



Results: Stanford2D3DS

Method	Configuration / Modalities	mIoU %	mAcc %	# Params (M)
CAT-Seg	Open-vocabulary with RGB	39.60	–	59.5
OpenSeeD		40.00	–	65.4
OOOPS		42.60	–	8.7
SAM3		52.05	65.02	850
SFSS-MMSI		52.87	63.96	40
HoHoNet	RGB	51.99	62.97	70
PanoFormer		52.35	64.31	20
CBFC		52.20	<u>65.60</u>	–
Tangent		45.60	65.20	–
Trans4PASS+		52.04	63.98	39
MultiPanoWise		<u>54.60</u>	–	–
PanoSAMic (ours)	RGB	59.62	74.11	178
SFSS-MMSI	RGB-D	55.49	68.57	81
HoHoNet		56.73	68.23	70
PanoFormer		<u>57.03</u>	68.08	20
CBFC		56.70	<u>70.80</u>	–
Tangent		52.50	70.10	–
360BEV		54.30	–	27.7
PanoSAMic (ours)	RGB-D	60.90	73.95	184
SFSS-MMSI	RGB-D-N	<u>59.43</u>	<u>69.03</u>	123
PanoSAMic (ours)	RGB-D-N	61.57	74.04	191

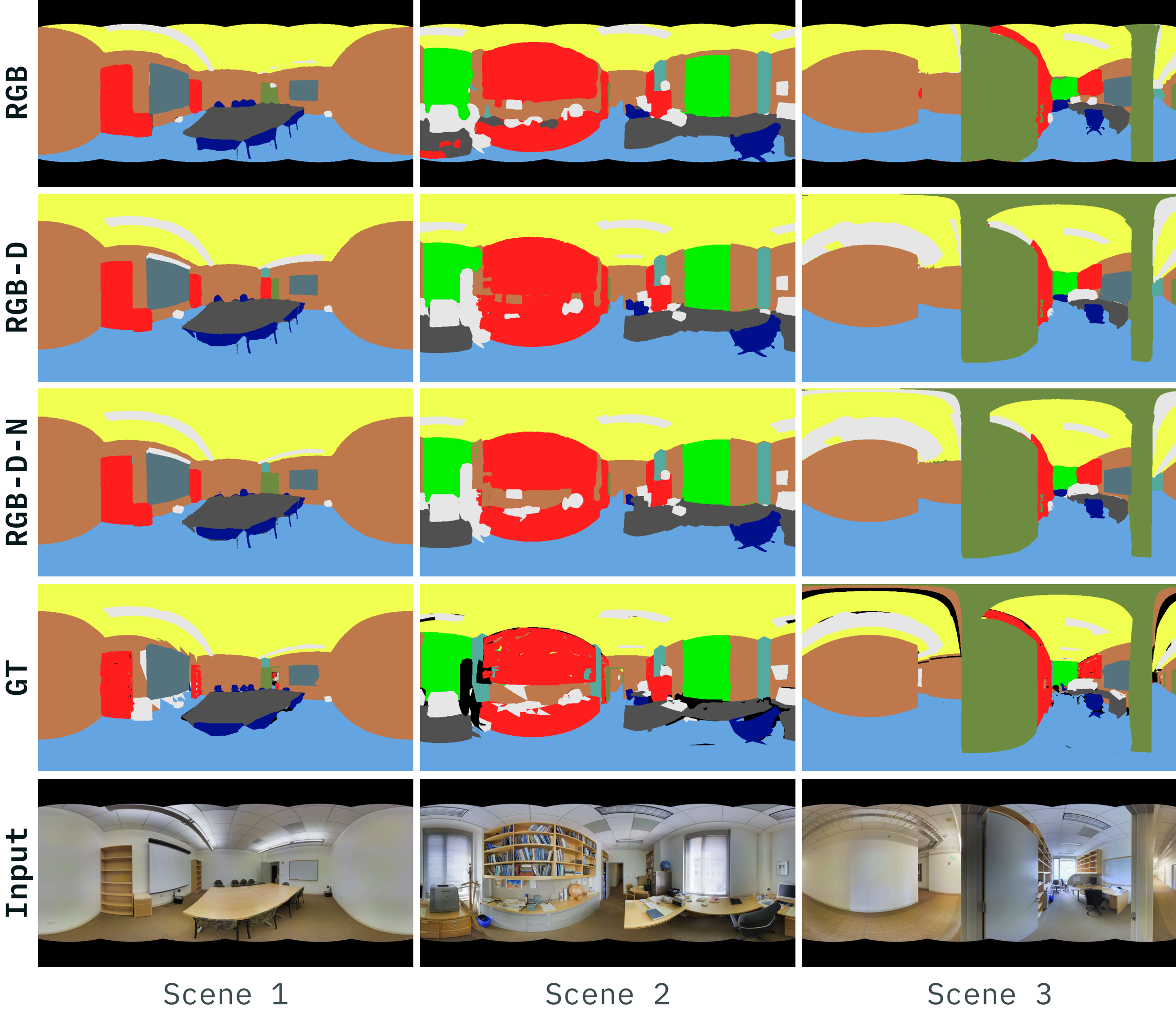
3-fold validation. RGB → RGB-D-N costs only +13 M parameters.

Results: Matterport3D

Method	Configuration / Modalities	mIoU %
CAT-Seg	Open-vocabulary with RGB	31.10
OpenSeeD		31.60
OOOPS		32.50
SAM3		42.29
Trans4PASS+		42.60
HoHoNet	RGB	44.10
SegFormer		45.53
PanoSAMic (ours)	RGB	46.59
360BEV	RGB-D	<u>46.35</u>
PanoSAMic (ours)	RGB-D	48.43

Splits and pre-processing follow 360BEV.

Qualitative Results



RGB degrades with distance; depth adds detail but sometimes reads walls as bookcases; RGB-D-N is most faithful near and far.

Generalization



On this outdoor scene, absent from indoor training, PanoSAMic labels more completely than the annotation.

Conclusion

- A frozen SAM encoder suffices: branch outputs plus a sphere-aware decoder reach **61.57 % mIoU**, backbone untouched.
- The gain comes from the panoramic components, not the backbone: a plain SAM encoder with a convolutional decoder reaches **53.29 % mIoU**, while the full model reaches **62.43 %** on Area 5 (**+9.14**).
- Dual-view fusion turns the seam into redundancy: edge mIoU *rises* (**55.52 → 60.39**) where single view falls.
- Attention must be local for dense prediction: MCBAM beats global CBAM; instance refinement adds **+0.82 mIoU**.
- Some of the lowest-scoring samples coincide with imprecise annotations, so panoramic benchmarks may under-report performance.